

go run .\priority_inbox\main.go

Campus Notification Platform - Priority Inbox

Fetching notifications from: http://20.207.122.201/evaluation-service/notifications

error fetching notifications: API returned status 401: {"message":"invalid authorization token"}
⚠ Could not reach API - running with sample data for demonstration.

Total notifications fetched: 14

TOP 10 PRIORITY NOTIFICATIONS

Score = TypeWeight + RecencyScore

Weights: Placement=300 Result=200 Event=100

Recency: $\max(0, 100 - \text{floor}(\text{hours_since_notification}))$

- #01 Placement Score: 400 (type=300 + recency=100)
ID: b283218f-ea5a-4b7c-93a9-1f2f240d64b0
Message: CSX Corporation hiring
Time: 2026-05-02 05:50:54 UTC (0.1h ago)
- #02 Placement Score: 400 (type=300 + recency=100)
ID: 8a7412bd-6065-4d09-8501-a37f11cc848b
Message: Advanced Micro Devices Inc. hiring
Time: 2026-05-02 05:40:54 UTC (0.3h ago)
- #03 Placement Score: 400 (type=300 + recency=100)
ID: d1a2b3c4-0001-0001-0001-000000000002
Message: Microsoft hiring - backend engineer
Time: 2026-05-02 05:35:54 UTC (0.3h ago)
- #04 Placement Score: 400 (type=300 + recency=100)
ID: d1a2b3c4-0001-0001-0001-000000000001
Message: Google hiring - software engineer
Time: 2026-05-02 05:45:54 UTC (0.2h ago)
- #05 Result Score: 300 (type=200 + recency=100)
ID: 0005513a-142b-4bbc-8678-eefec65e1ede
Message: mid-sem

```
#05 Result      Score: 300 (type=200 + recency=100)
    ID:      0005513a-142b-4bbc-8678-eefec65e1ede
    Message: mid-sem
    Time:    2026-05-02 05:25:54 UTC (0.5h ago)

#06 Result      Score: 300 (type=200 + recency=100)
    ID:      ea836726-c25e-4f21-a72f-544a6af8a37f
    Message: project-review
    Time:    2026-05-02 05:03:54 UTC (0.9h ago)

#07 Result      Score: 299 (type=200 + recency= 99)
    ID:      e5c4ff20-31bf-4d40-8f02-72fda59e8918
    Message: project-review
    Time:    2026-05-02 04:37:54 UTC (1.3h ago)

#08 Result      Score: 299 (type=200 + recency= 99)
    ID:      003cb427-8fc6-47f7-bb00-be228f6b0d2c
    Message: external
    Time:    2026-05-02 04:51:54 UTC (1.1h ago)

#09 Result      Score: 299 (type=200 + recency= 99)
    ID:      d1a2b3c4-0001-0001-0001-000000000004
    Message: end-semester results published
    Time:    2026-05-02 04:55:54 UTC (1.0h ago)

#10 Result      Score: 297 (type=200 + recency= 97)
    ID:      cf2885a6-45ac-4ba0-b548-6e9e9d4c52c8
    Message: project-review
    Time:    2026-05-02 02:55:54 UTC (3.0h ago)
```

Scoring summary

Placement: 4 Result: 6 Event: 4 Total scored: 14

Heap algorithm

Min-heap capacity: 10

Notifications processed: 14

Time complexity: $O(M \log N)$ where $M=14$, $N=10$

Each new notification: $O(\log N)$ insertion - efficient for live streams

go run .\vehicle_scheduling\scheduler.go

```
Vehicle Maintenance Scheduler - 0/1 Knapsack DP

Mechanic-hour budget: 10.0 hours
Algorithm: 0/1 Knapsack DP - O(N * W) time

--- Step 1: Fetching depots ---
⚠ API unavailable (API /depots returned status 401: {"message":"invalid authorization token"})
Using sample depot data.

Depot: depot-001 - Mumbai Central Depot
Depot: depot-002 - Pune East Depot

--- Step 2: Fetching maintenance tasks ---
⚠ API unavailable (API /depots/depot-001/tasks returned status 404: 404 page not found)
Using sample task data.

Total tasks available: 15

TaskID  Duration  Score  Description
-----
t-01    1.0h     60    Oil change & filter replacement
t-02    2.0h    100    Brake inspection & pad replacement
t-03    3.5h    150    Engine diagnostic & tune-up
t-04    1.5h     80    Tyre rotation & balancing
t-05    2.5h    120    Transmission fluid flush
t-06    4.0h    160    Full body inspection
t-07    0.5h     40    Battery replacement
t-08    2.0h     90    Coolant system flush
t-09    0.5h     30    Air filter replacement
t-10    3.0h    140    Suspension & steering check
t-11    2.5h    110    Exhaust system repair
t-12    1.5h     70    AC system service
t-13    1.0h     50    Headlight & electrical check
t-14    2.0h     95    Fuel injector cleaning
t-15    0.5h     20    Windshield wiper replacement

--- Step 3: Running 0/1 Knapsack DP ---
N = 15 tasks, W = 10.0h budget (100 scaled units)

--- Step 3: Running 0/1 Knapsack DP ---
N = 15 tasks, W = 10.0h budget (100 scaled units)

OPTIMAL MAINTENANCE SCHEDULE

Budget:          10.0 hours
Total duration:  10.0 hours (100.0% utilisation)
Remaining:       0.0 hours
Total score:     525 (maximum achievable)

TaskID  Duration  Score  Description
-----
t-05    2.5h    120    Transmission fluid flush
t-02    2.0h    100    Brake inspection & pad replacement
t-14    2.0h     95    Fuel injector cleaning
t-04    1.5h     80    Tyre rotation & balancing
t-01    1.0h     60    Oil change & filter replacement
t-07    0.5h     40    Battery replacement
t-09    0.5h     30    Air filter replacement

--- Complexity analysis ---
Time: O(N * W) = O(15 * 100) = O(1500) operations
Space: O(N * W) = O(1616) DP table entries
Suitable for real-world scale: N=1000 tasks, W=480h budget → ~480K ops
```

go run .\vehicle_maintenance_scheduler\main.go

POST

▼

http://localhost:8080/api/v1/vehicles

ParamsAuthorizationHeaders (8)Body●Pre-request ScriptTestsSettings

☐ none

☐ form-data

☐ x-www-form-urlencoded

☒ raw

☐ binary

JSON▼

1

{

2

"owner_name": "Rahul Sharma",

3

"registration_no": "MH12AB1234",

4

"make": "Maruti",

5

"model": "Swift",

6

"year": 2022

7

}

BodyCookiesHeaders (3)Test Results

PrettyRawPreviewVisualize

JSON▼

1

{

2

"id": "17a1b95a-1b5f-307a-02b2-87ab6f13886b",

3

"owner_name": "Rahul Sharma",

4

"registration_no": "MH12AB1234",

5

"make": "Maruti",

6

"model": "Swift",

7

"year": 2022,

8

"created_at": "2026-05-02T06:05:46.9830102Z"

9

}



http://localhost:8080/api/v1/vehicles

GET



http://localhost:8080/api/v1/vehicles

Params

Authorization

Headers (6)

Body

Pre-request Script

Tests

Settings

Query Params

	Key	Value
	Key	Value

Body

Cookies

Headers (3)

Test Results

Pretty

Raw

Preview

Visualize

JSON



```
1  [
2    {
3      "id": "17a1b95a-1b5f-307a-02b2-87ab6f13886b",
4      "owner_name": "Rahul Sharma",
5      "registration_no": "MH12AB1234",
6      "make": "Maruti",
7      "model": "Swift",
8      "year": 2022,
9      "created_at": "2026-05-02T06:05:46.9830102Z"
10   }
11 ]
```

POST

▼

http://localhost:8080/api/v1/jobs

ParamsAuthorizationHeaders (8)Body●Pre-request ScriptTestsSettings

none

form-data

x-www-form-urlencoded

raw

binary

JSON

▼

1

}

2

"vehicle_id": "17a1b95a-1b5f-307a-02b2-87ab6f13886b",

3

"type": "oil_change",

4

"scheduled_at": "2026-06-10T10:00:00Z",

5

"notes": "Use synthetic oil"

6

}

BodyCookiesHeaders (3)Test Results

PrettyRawPreviewVisualizeJSON▼

1

{

2

"id": "b3f6806f-2a33-7ac2-039d-2cb730058791",

3

"vehicle_id": "17a1b95a-1b5f-307a-02b2-87ab6f13886b",

4

"type": "oil_change",

5

"status": "scheduled",

6

"scheduled_at": "2026-06-10T10:00:00Z",

7

"notes": "Use synthetic oil",

8

"created_at": "2026-05-02T06:06:57.9861131Z",

9

"updated_at": "2026-05-02T06:06:57.9861131Z"

10

}

GET

⌵

http://localhost:8080/api/v1/jobs/b3f6806f-2a33-7ac2-039d-2cb730058791

Params

Authorization

Headers (6)

Body

Pre-request Script

Tests

Settings

Query Params

	Key	Value
	Key	Value

Body

Cookies

Headers (3)

Test Results

 Status: 200 OK 1

Pretty

Raw

Preview

Visualize

JSON ⌵



```
1  {
2    "id": "b3f6806f-2a33-7ac2-039d-2cb730058791",
3    "vehicle_id": "17a1b95a-1b5f-307a-02b2-87ab6f13886b",
4    "type": "oil_change",
5    "status": "scheduled",
6    "scheduled_at": "2026-06-10T10:00:00Z",
7    "notes": "Use synthetic oil",
8    "created_at": "2026-05-02T06:06:57.9861131Z",
9    "updated_at": "2026-05-02T06:06:57.9861131Z"
10 }
```

PATCH

http://localhost:8080/api/v1/jobs/b3f6806f-2a33-7ac2-039d-2cb730058791/status ...

Params

Authorization

Headers (8)

Body

Pre-request Script

Tests

Settings

none

form-data

x-www-form-urlencoded

raw

binary

JSON

1

2

3

4

5

```
"status": "in_progress",
"technician_name": "Amit Patil",
"notes": "Started oil change"
```

Body

Cookies

Headers (3)

Test Results

Status: 200 OK

Pretty

Raw

Preview

Visualize

JSON

1

2

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11

```
{
  "id": "b3f6806f-2a33-7ac2-039d-2cb730058791",
  "vehicle_id": "17a1b95a-1b5f-307a-02b2-87ab6f13886b",
  "type": "oil_change",
  "status": "in_progress",
  "scheduled_at": "2026-06-10T10:00:00Z",
  "notes": "Started oil change",
  "technician_name": "Amit Patil",
  "created_at": "2026-05-02T06:06:57.9861131Z",
  "updated_at": "2026-05-02T06:08:06.7963969Z"
}
```

GET

⌵

http://localhost:8080/api/v1/vehicles/17a1b95a-1b5f-307a-02b2-87ab6f13886b/jobs ...

Params

Authorization

Headers (6)

Body

Pre-request Script

Tests

Settings

Query Params

	Key	Value
	Key	Value

Body

Cookies

Headers (3)

Test Results

Pretty

Raw

Preview

Visualize

JSON ⌵

```
1  [
2    {
3      "id": "b3f6806f-2a33-7ac2-039d-2cb730058791",
4      "vehicle_id": "17a1b95a-1b5f-307a-02b2-87ab6f13886b",
5      "type": "oil_change",
6      "status": "in_progress",
7      "scheduled_at": "2026-06-10T10:00:00Z",
8      "notes": "Started oil change",
9      "technician_name": "Amit Patil",
10     "created_at": "2026-05-02T06:06:57.9861131Z",
11     "updated_at": "2026-05-02T06:08:06.7963969Z"
12   }
13 ]
```

go run .\notification_app_be\main.go

GET http://localhost:8081/health

Params Authorization Headers (6) Body Pre-request Script Tests Settings

Query Params

Key	Value
Key	Value

Body Cookies Headers (3) Test Results  Status: 200 OK

Pretty Raw Preview Visualize JSON 

```
1 {
2   "service": "notification-app-be",
3   "status": "ok"
4 }
```

 http://localhost:8081/api/v1/notifications

POST http://localhost:8081/api/v1/notifications

Params Authorization Headers (8) Body Pre-request Script Tests Settings

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary JSON 

```
2 {
3   "job_id": "job-001",
4   "recipient_id": "student-42",
5   "channel": "email",
6   "subject": "Placement Drive",
7   "body": "Google is hiring - attend the drive on June 10"
8 }
```

Body Cookies Headers (3) Test Results  Status: 201 Created Time: 17 ms

Pretty Raw Preview Visualize JSON 

```
1 {
2   "id": "09fa9bdf-24a9-882e-7c82-441e1e81cac5",
3   "job_id": "job-001",
4   "recipient_id": "student-42",
5   "channel": "email",
6   "subject": "Placement Drive",
7   "body": "Google is hiring - attend the drive on June 10",
8   "status": "sent",
9   "attempts": 0,
10  "sent_at": "2026-05-02T06:38:35.4926271Z",
11  "created_at": "2026-05-02T06:38:35.4926271Z",
12  "updated_at": "2026-05-02T06:38:35.4926271Z"
13 }
```

GET

⌵

http://localhost:8081/api/v1/notifications

Params

Authorization

Headers (6)

Body

Pre-request Script

Tests

Settings

Query Params

	Key	Value
	Key	Value

Body

Cookies

Headers (3)

Test Results

🌐

Status: 200 OK

Pretty

Raw

Preview

Visualize

JSON ⌵

⌵

```
1  [
2    {
3      "id": "09fa9bdf-24a9-882e-7c82-441e1e81cac5",
4      "job_id": "job-001",
5      "recipient_id": "student-42",
6      "channel": "email",
7      "subject": "Placement Drive",
8      "body": "Google is hiring – attend the drive on June 10",
9      "status": "sent",
10     "attempts": 0,
11     "sent_at": "2026-05-02T06:38:35.4926271Z",
12     "created_at": "2026-05-02T06:38:35.4926271Z",
13     "updated_at": "2026-05-02T06:38:35.4926271Z"
14   }
15 ]
```

GET

⌵

http://localhost:8081/api/v1/notifications/09fa9bdf-24a9-882e-7c82-441e1e81cac5

Params

Authorization

Headers (6)

Body

Pre-request Script

Tests

Settings

Query Params

	Key	Value
	Key	Value

Body

Cookies

Headers (3)

Test Results

⌐

Status: 200 OK

Pretty

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Preview

Visualize

JSON ⌵

⌵

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⌵

"id": "09fa9bdf-24a9-882e-7c82-441e1e81cac5",

"job_id": "job-001",

"recipient_id": "student-42",

"channel": "email",

"subject": "Placement Drive",

"body": "Google is hiring – attend the drive on June 10",

"status": "sent",

"attempts": 0,

"sent_at": "2026-05-02T06:38:35.4926271Z",

"created_at": "2026-05-02T06:38:35.4926271Z",

"updated_at": "2026-05-02T06:38:35.4926271Z"

⌵